KL-VAE: Variational Autoencoders with KL Divergence

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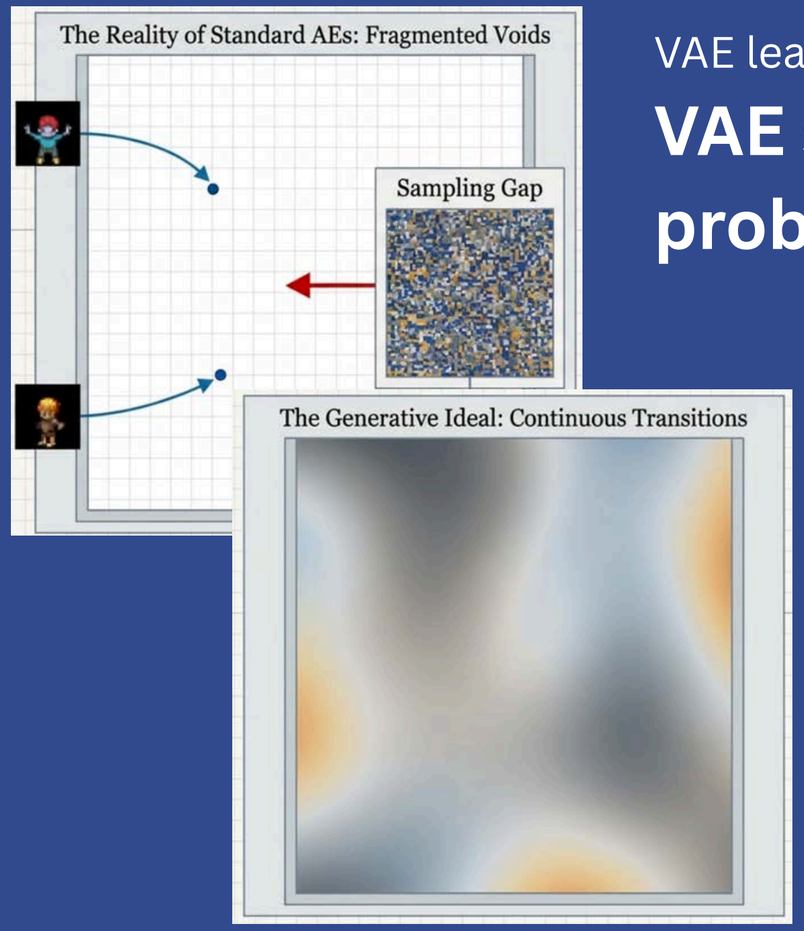
course: kecerdasan buatan generatif pada visi komputer

What is a Variational Autoencoder (VAE)?



VAE learn to encode data to latent space like GAN

VAE solves the gap by mapping images to probability distribution instead of single points

Encoder

Decoder

Image Input → Latent Distribution → Reconstruct Output

Continuous
normally distributed

Reconstruction Loss

Minimize difference between input and output

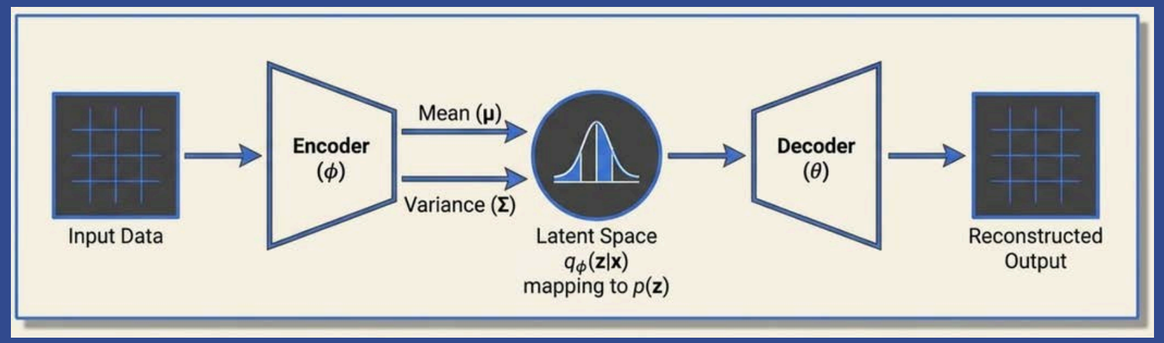
VS

Kullback-Leibler Divergence

Force distribution to match a standard normal distribution

VAE vs GAN Comparison

Variational Autoencoder



Compress data into a **probabilistic** latent space

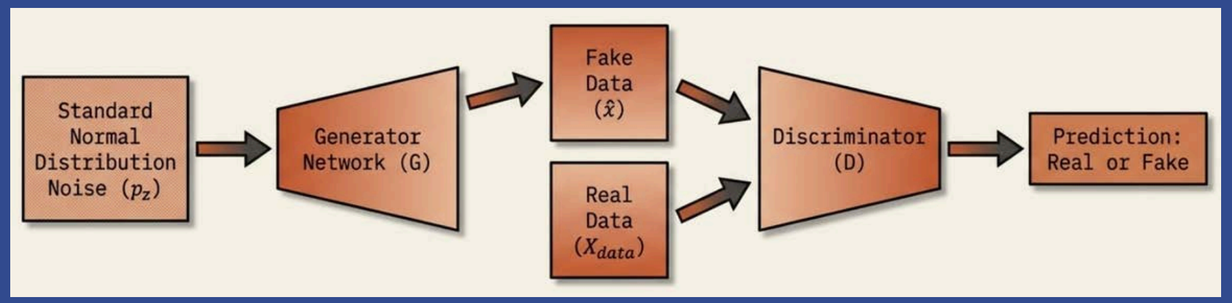
Generate data from the sample of the distribution

Goal: Maximize Evidence Lower Bound (ELBO)

Generative Adversarial Network

Two neural network compete to generate and differentiate fake data

Generate data from a random noise



Goal: Reach Nash Equilibrium

VAE vs GAN Comparison

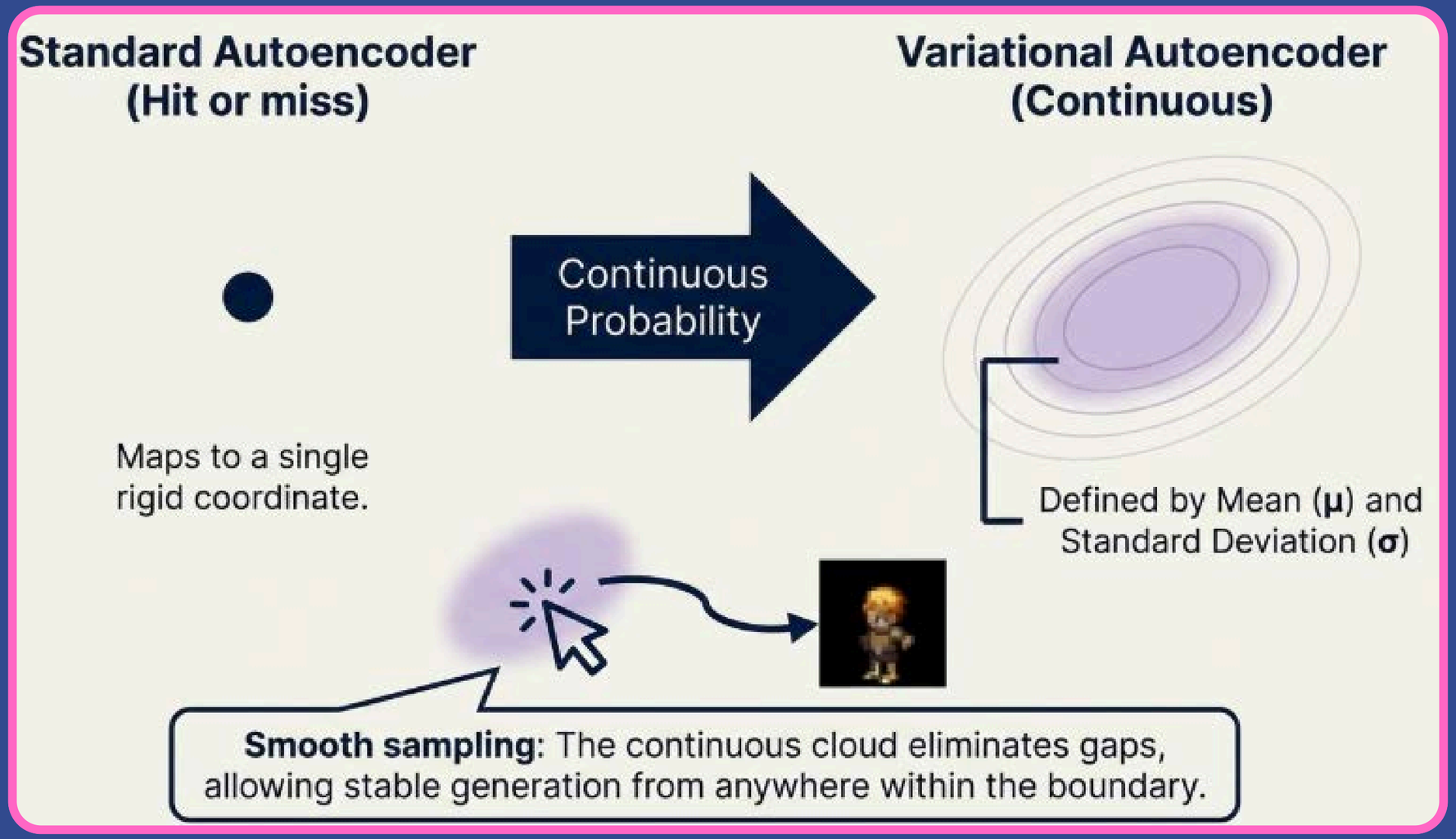
VAE is BLURRY

VAE is penalized by MSE.
MSE is minimized the average error
Thus the image is blurry

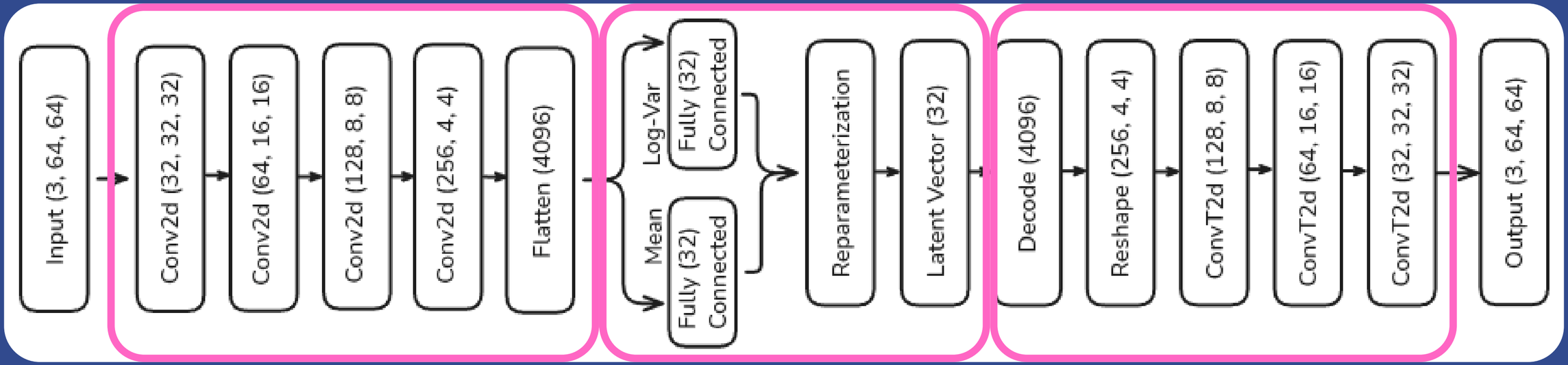
GAN is SHARP

GAN is penalized by the discriminator
Sharp but different image is better
Thus the image is sharp

Standard Autoencoder vs VAE



KL-VAE Architecture Overview



Encoder

Latent Vectors

Decoder

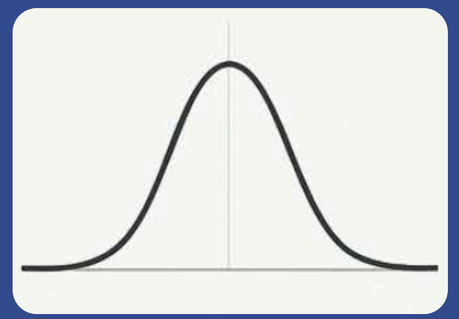
VAE Loss Function

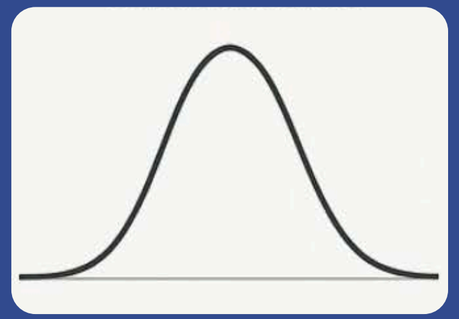
Reconstruction Loss

How different are these

Train decoder to recreate input accurately

KL Divergence





How far apart are these

Encourage smooth, continuous latent space

Total Loss = Reconstruction Loss + KL Divergence

Understanding KL Divergence (Kullback–Leibler)

What is KL-Divergence

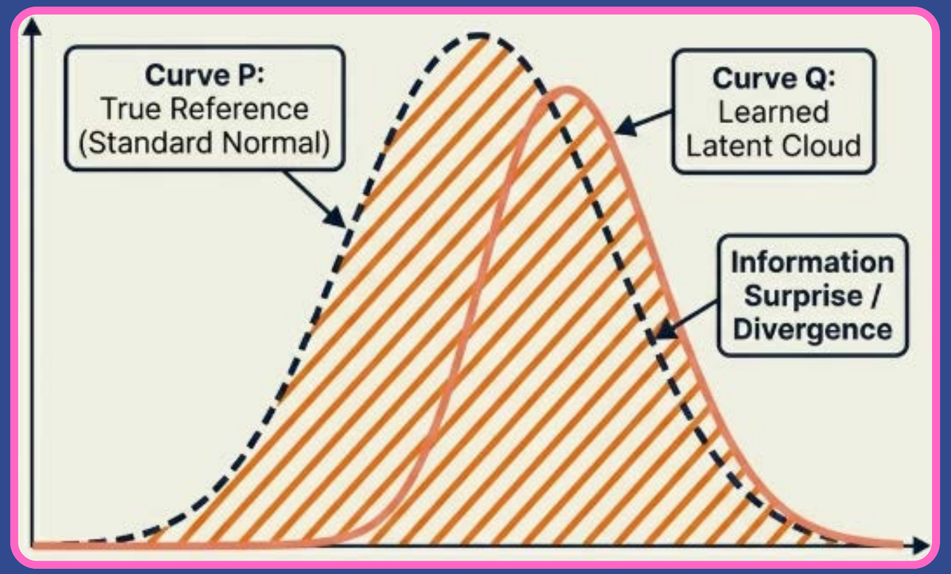
Measure of how one probability distribution differs from one another

Equals to 0 only if P and Q are identical

KL-Divergence in VAE context

Encourages latent distribution to be simple

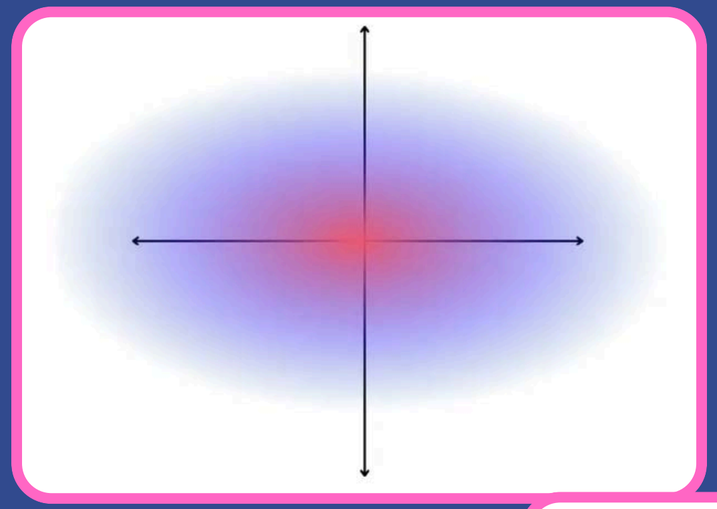
Prevent model from encoding all information without compression



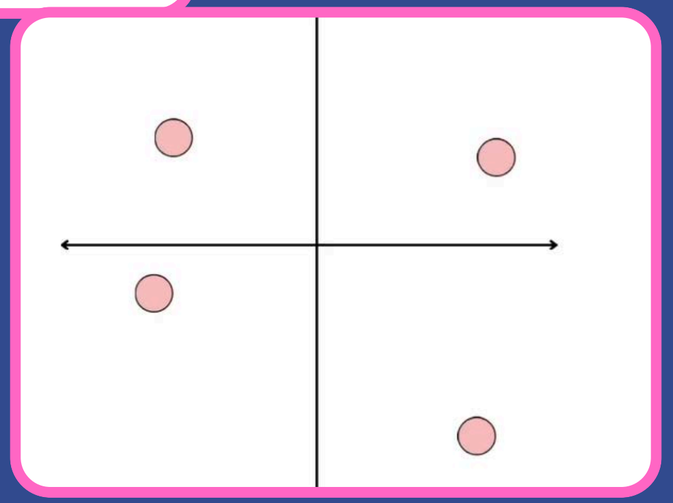
Small KL Loss → Latent space close to normal dist → Better sampling
Large reconstruction loss → Good encoding → Perfect reconstruction
Balance between two creates meaningful representation

KL-VAE vs VQ-VAE Comparison

Property	KL-VAE	VQ-VAE
Latent Space	Continuous [32]	Discrete [512]
Distribution	Gaussian	Empirical
Interpolation	Smooth	Discrete
Interpretability	High (Latent Dims)	High (Codebook)
Use Case	Interpolation	Compression



Continuous Latent Space



Discrete Latent Space

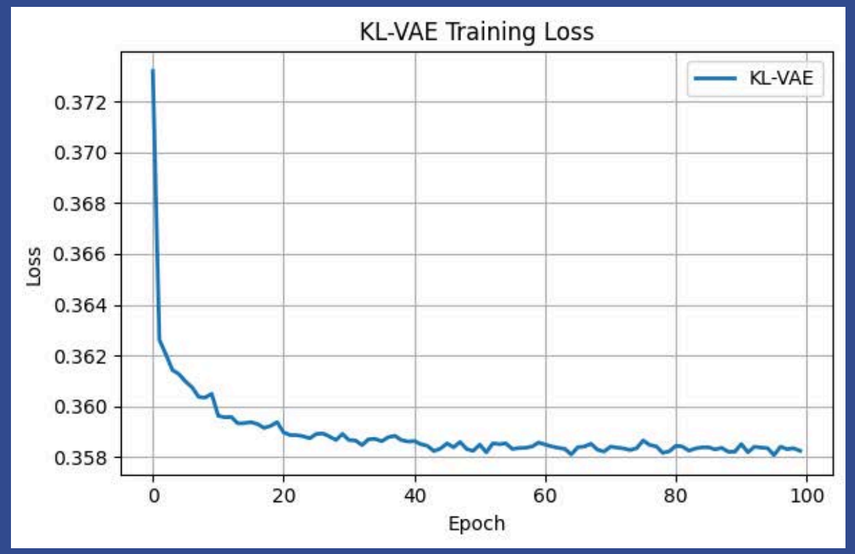
VQ-VAE - Better for reconstruction
KL-VAE - Better for Sampling

Training Setup

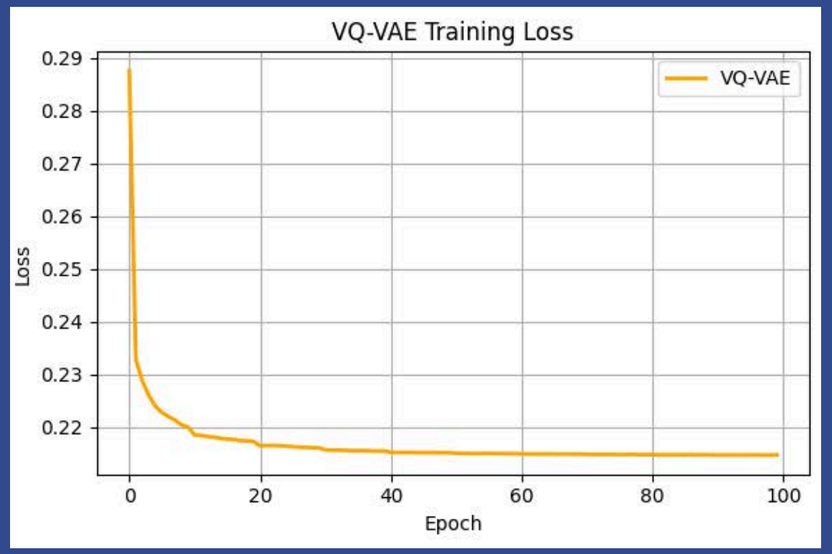
Params	Value
Train:Test	80:20
Batch Size	32
Epochs	100
Optimizer	Adam
Learning Rate	0.001
GPU	5070Ti

Anime Faces Dataset
63,565 Images

Training Results



No Oscillation



Good Convergence

Model	Train Time (m)	Final Loss	Params
KL-VAE	50.91	0.3583	1,777,411
VQ-VAE	54.69	0.2147	626,179

Reconstruction Quality (Quantitative)

Model	MSE	PSNR	Quality
KL-VAE	0.1935	13.16dB	Good
VQ-VAE	0.0785	17.07dB	Excellent

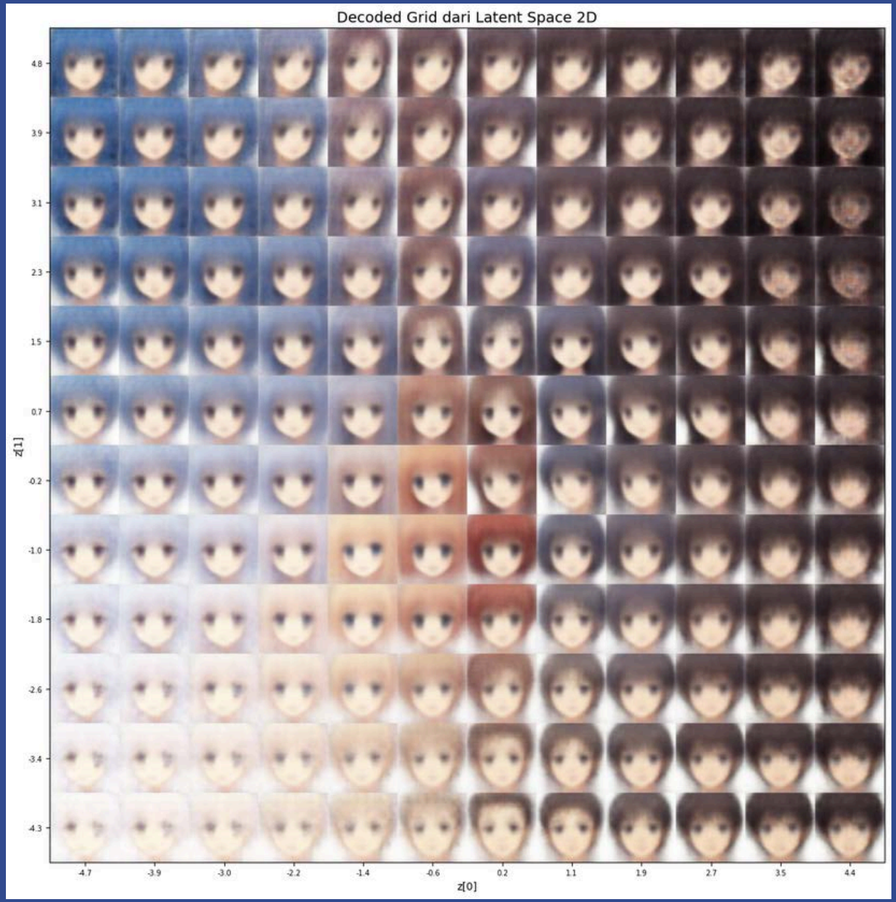
KL-VAE
Slightly blurry but natural looking faces

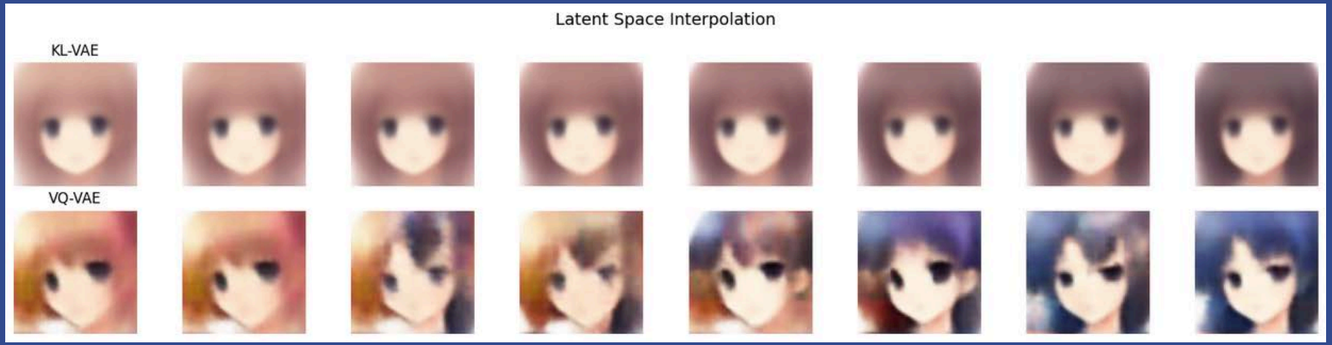
VQ-VAE
Sharp images with good detail preservation

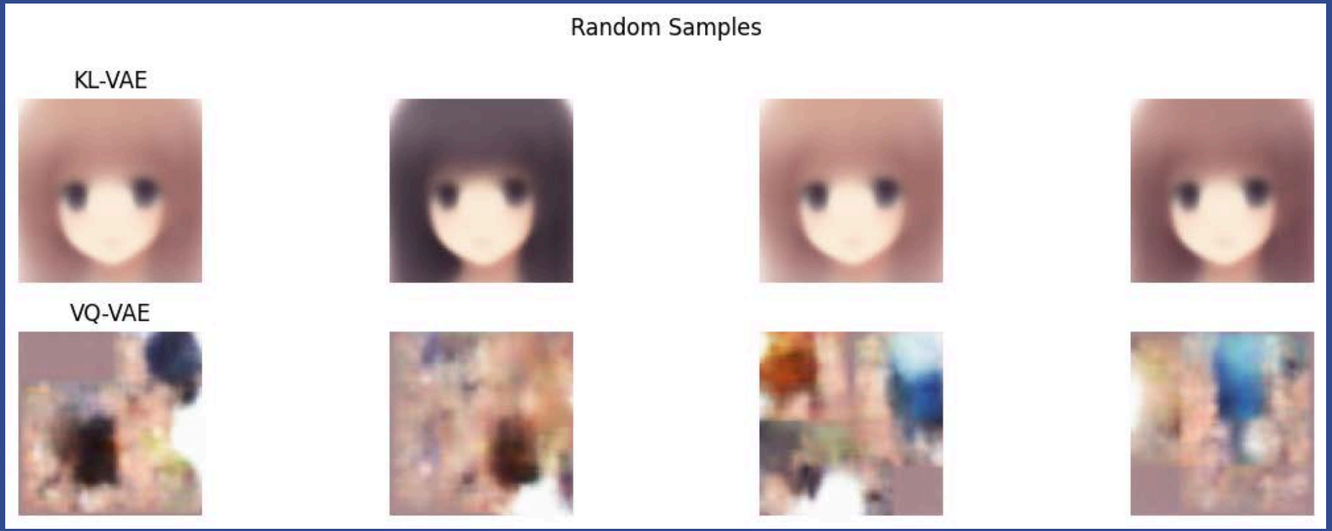
Discrete Quantization Helps Preserve Details

Continuous Space Scarifice Sharpness for Smootheness

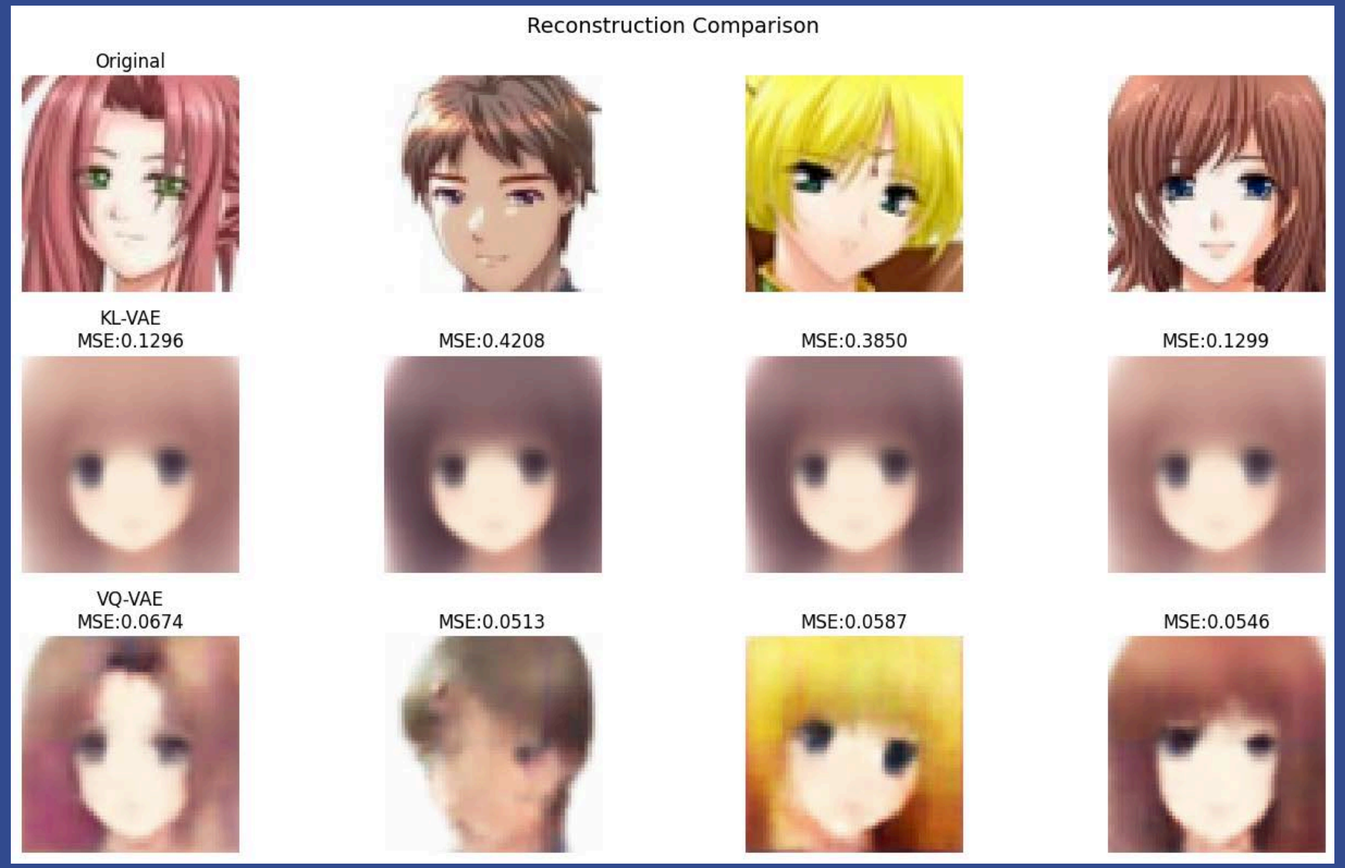
Reconstruction Quality (Qualitative)







Reconstruction Quality KL-VAE vs GAN



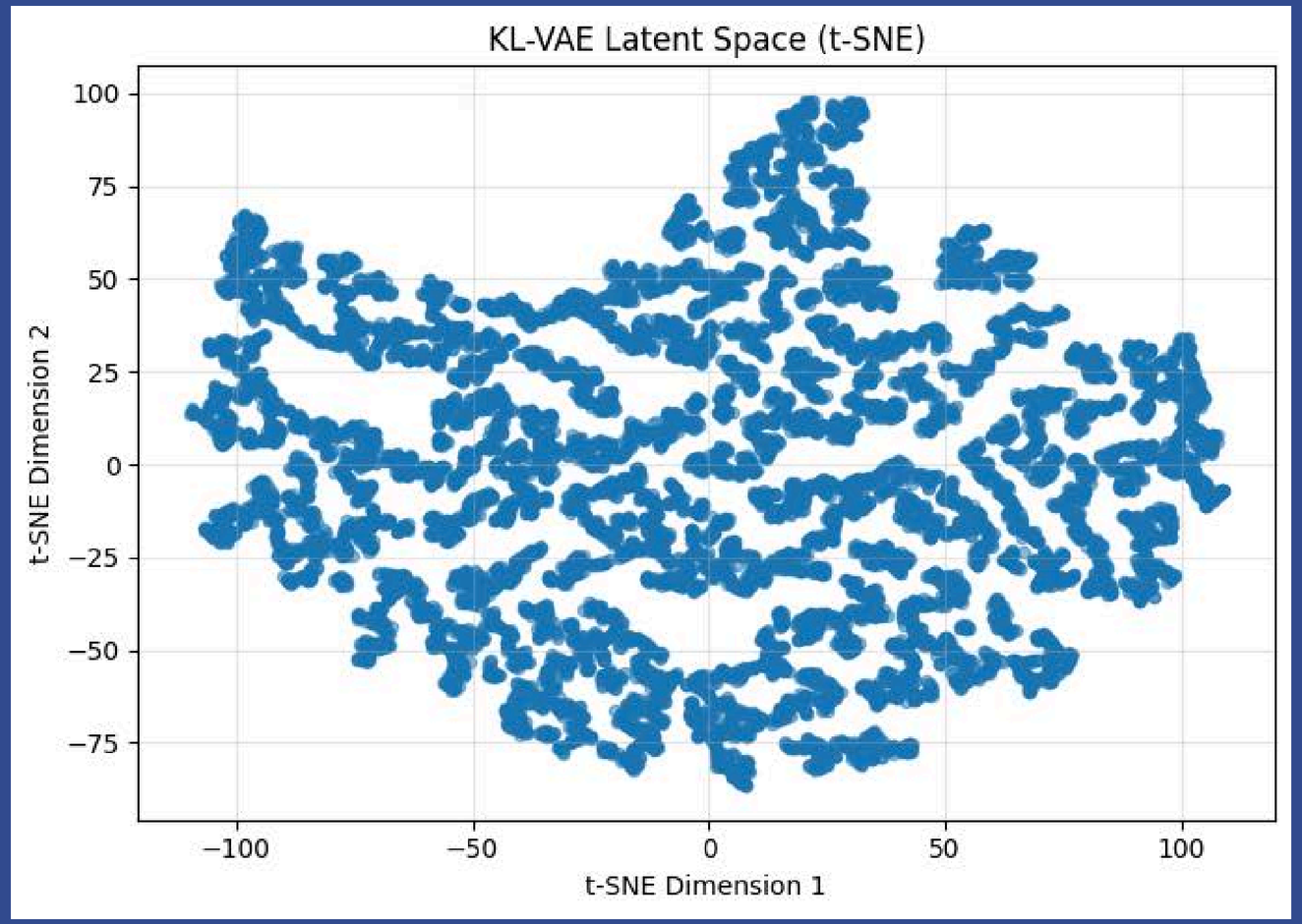
Latent Space Visualization

Test Images
1272

t-SNE Dimension Reduction
32D → 2D

Preplexity **30**

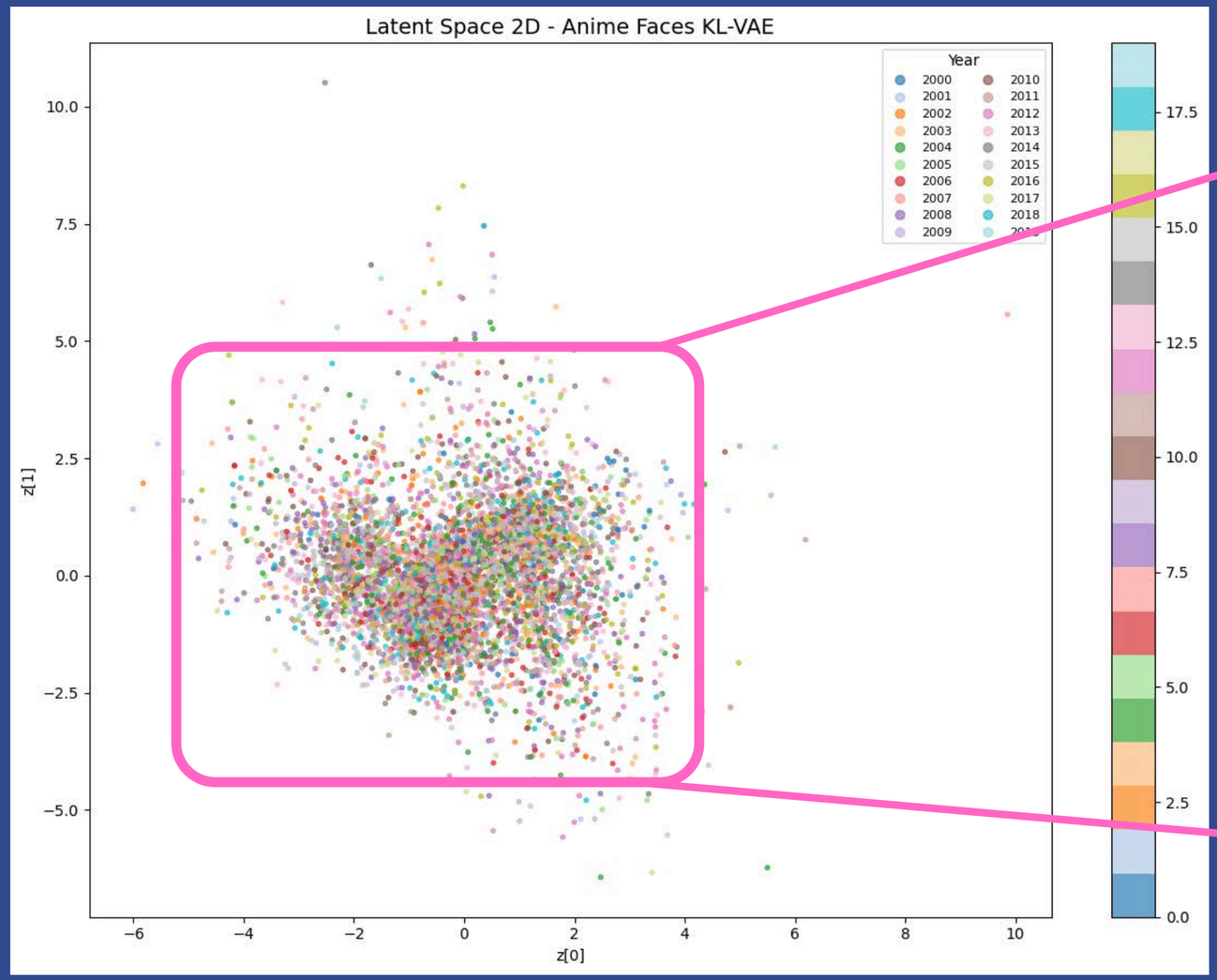
Iterations **1000**



Our Observations

- Clear clustering patterns emerge
- No distinct classes but natural grouping
- Smooth transition between clusters suggest good interpolation

Latent Space Visualization (class by year)



Conclusion

Key Takeaways

Reparameterization trick makes VAE trainable and practical

Trade-offs exist between reconstruction and regularity

KL-VAE & VQ-VAE (Continuous vs Discrete) Serve different purposes

VAE Applications

Image Compression and Transmission
Data Augmentation
Image Anomaly Detection
Style Transfer

Real world data across classes is not continuous and interpolating these classes might not make sense at all. Either the image is of a cat or a person, interpolating across classes cat and person does not generate a representative image.

Thank You, Any Question?

Scan Me

https://github.com/AbiyaMakruf/TelU-CaseBased-VAE-KecerdasanBuatanGeneratifPadaVisiKomputer